

ACL-Style Review: PD-Prune: Disaggregated Prefill–Decode Pruning for Efficient Agentic LLM Serving

Internal Meta-Review

October 8, 2026

Abstract

This document is an **ACL-style review** of the current draft: strengths, weaknesses, writing/notation issues, factual inconsistencies, and experiments that need strengthening. It also records a DistServe-style core architecture figure proposal (`fig/fig_core_arch.tex`).

Overall recommendation

Borderline / Major Revision (as a systems-for-NLP paper at ACL).

The core claim is clear and timely: under prefill–decode disaggregation, decode-time path pruning is too late because residual prefill and KV transfer have already been paid; admitting residuals on the prefill side of the cut (with a trunk alias) can cut work and connector traffic without changing the decode host. The formalization (trunk / residual / spine; W , X ; action set) is a genuine contribution.

The draft is not yet ACL-ready. The writing is telegraphic to the point of opacity; naming is inconsistent (PD-Prune/ FS / APP / PD-Prune+); the scoring functions that drive every threshold are underspecified; several experiments use tiny slices and a synthetic “loop + illegal” residual mix; and the main text / appendix largely duplicate each other. Fixing these is necessary before the paper can be fairly judged on technical merit.

Scores (ACL form, 1–5):

Soundness	3
Excitement / Novelty	3.5
Reproducibility	2
Clarity	2
Meaningful comparison	3
Overall assessment	2.5

1 Summary of the paper

Agentic / tree-search serving expands one committed *trunk* of length L into k *residuals* and keeps one *spine*. Prefix caching, PD-disaggregation, and decode-time pruners (ESC, Speculative Rejection, DPTS) each pay for non-winners before rejection is possible. PD-Prune places two independent gates on that fan-out:

- (a) **Radix prune** (prefill-side): first-match action $a_i \in \{\text{HS}, \text{HP}, \text{DS}, \text{EA}, \text{PF}\}$ from page match m_i , loop set Λ , and early-abort bit F_i ; trunk is CoW-aliased so $L \notin W_i$.

(b) **Decode prune:** probe of t_0 tokens; continue to D iff $s'_i \geq \tau_{\text{dec}}$. Defaults $(\tau_{\text{pre}}, \tau_{\text{dec}}) = (0.15, 0.45)$.

Reported wins: -23% computed prefill on GSM8K at matched any-finished; MATH-500 probe restores 20/32 vs. blanket drop 13/32; spine peak KV $\approx 0.72 \times$ APC.

2 Strengths

1. **Right problem for agentic serving.** Moving the prune decision before the PD connector is a clean systems insight and is well motivated against DistServe / Splitwise and decode-time pruners.
2. **Useful abstraction.** Trunk / residual / spine and the pair (W, X) make the “who pays when” argument precise; Lemma 1 (first-fan-out dominance) is the right kind of claim.
3. **Host-invariance** (Prop. 2) is a strong experimental design idea: one τ_{pre} moves the prefill column for ESC / SR / DPTS.
4. **Probe vs. blanket drop** on MATH-500 is a good negative result that justifies threshold separation.
5. **Figures** (`fig_pd`, `fig_prune`, `fig_trad`) already communicate the cut better than much of the prose.

3 Major weaknesses

3.1 W1. Venue fit and audience

This is primarily an **inference-systems** paper (KV pages, connectors, CoW page tables, goodput). ACL readers will ask: what is the NLP contribution beyond plugging existing pruners into a serving stack? The draft never states a crisp NLP takeaway (e.g., how admission changes *which* reasoning traces survive, or whether score design interacts with math / agent semantics). Either (i) foreground the NLP/agent consequences and evaluate on real agent traces, or (ii) target a systems venue (OSDI/NSDI/EuroSys/MLSys) with the current framing.

3.2 W2. Scoring functions are undefined (blocks reproducibility)

Every operating point rests on $\text{Score}_{\text{pre}}$, $\text{Score}_{\text{dec}}$, and $\text{Loop}(\cdot)$. The paper only reports four draft numbers (0.02/0.22/1.00 and 0.04/0.16/0.92) and treats Prop. 1 as if these were general separators. Missing:

- Is the score a classifier, entropy, reward model, heuristic regex, or STOP-style learned token?
- Training / calibration data; false-positive rate of Loop .
- Latency of scoring (critical if it sits on the TTFT path).
- Sensitivity: what happens when score noise moves illegal text below τ_{pre} ?

Without this, the thresholds (0.15, 0.45) look hand-tuned to one synthetic mix, not a transferable method.

3.3 W3. Evaluation slices are too small / too synthetic

- Main GSM8K gate: $n=16$; MATH-500: first 32 level- ≥ 4 problems.
- Residuals are an artificial “hopeless mix” (contest strategies + repeated loops + illegal text). This maximizes the benefit of DS and may not match real ToT / ReAct fan-outs.

- Claims cite SWE-agent / OpenHands / ReAct, but GPU results are almost all math ToT; HumanEval is $k=2$ and barely exercises the gate.
- Control-plane microbench (tab:app style; $12\ \mu\text{s}/\text{token}$) is mixed into the narrative as if it were wall-clock evidence.

Needed: full GSM8K / MATH-500 under the gate; at least one real multi-tool agent workload; report $\text{mean} \pm \text{std}$ over seeds; separate synthetic model from GPU tables.

3.4 W4. Naming and system-variant confusion

The draft mixes **PD-Prune**, **FS**, **APP**, and **PD-Prune+** without a single variant table. Readers cannot tell whether a row is “alias only”, “alias + radix prune”, “+ decode probe”, or “+ last-line stop” (the latter is commented out in places but still appears in appendix tables). This hurts every comparison claim.

3.5 W5. Clarity / redundancy

- Abstract is unreadable without the body (“cutoffs are $(0.15, 0.45)$ ” with no score definition).
- Main Formulation / Method largely restate Appendix Background / Design.
- Intro dumps symbols, thresholds, and table pointers in one paragraph (“This work”).
- Prose style is telegram-dense (“Index 0 stays in $\mathcal{K}$ ”) without teaching the reader first—unfriendly for ACL.

4 Writing: unprofessional, redundant, or unclear

4.1 Redundancy (cut or demote)

Repeated material	Suggestion
Intro + Formulation + Method all restate “APC / PD / decode prune pay before reject”	Keep one crisp statement in Intro; cite Appendix for history.
Method §?? vs. Design §?? (same m_i, Λ, F_i, W, X)	Main: definitions + Alg. 1; Appendix: CoW / API / pool.
Results vs. Full Evaluation (same GSM8K / MATH numbers retold)	Main: one summary table + 2–3 figures; Appendix: ablations only.
“Reading one cold fan-out” paragraph repeats Tab. scores + Prop. 1	Keep one walkthrough; delete the second.
Related Work (main) vs. Extended Related Work	Main: 1 page; move taxonomy to Appendix.

4.2 Unprofessional / ACL-unfriendly phrasing

- Abstract opener: “Agentic serving expands one trunk...”—define *trunk* or avoid jargon in the first sentence.
- “The cutoffs are $(0.15, 0.45)$.”—sounds like a lab note, not a paper.
- “FS is the alias and PD-Prune+ is the same system with the admission rule enabled.”—define once in a boxed variant table.

- Overuse of imperative / elliptical sentences without subjects (“Abort leaves the spine.”). Fine in systems papers; heavy for ACL.
- Mixing code identifiers (`fork`, `insert`) into claims without an API subsection in the main text.

4.3 Suggested rewrite targets (priority)

1. Rewrite **Abstract** as: problem $\rightarrow$ method (2 sentences) $\rightarrow$ 3 quantitative results with setting.
2. Rewrite **Intro** around one figure + three bullets; move thresholds into Method.
3. Add a “**What is scored?**” subsection before Alg. 1.
4. Collapse main Experiments to ≤ 2 pages; push forests / sweeps to Appendix.

5 Errors and inconsistencies

5.1 E1. Algorithm vs. equations (transfer predicate)

Alg. app sets

$$X_i \leftarrow W_i [i=0 \vee a_i \neq \text{DS}] [a_i \in \{\text{HP}, \text{PF}\}],$$

while eq:xfer uses $[i \in \mathcal{K}]$. With $\mathcal{K} = \{0\} \cup \{i : a_i \neq \text{DS}\}$ these coincide *only if* EA is never transferred—which eq:xfer already enforces via the $\{\text{HP}, \text{PF}\}$ filter. Still, the algorithm should use the same predicate as the equation ($i \in \mathcal{K}$) to avoid reviewer nitpicks. Also: EA contributes $W_i = \lceil \alpha \ell_i \rceil$ but $X_i = 0$ —state explicitly that early-abort pages are dropped on the prefill side and never inserted.

5.2 E2. Score naming

Main text: $\text{Score}_{\text{pre}}, \text{Score}_{\text{dec}}$. Alg. probe: DraftScore . Design appendix: $s(\rho_i[0:\lceil \alpha \ell_i \rceil]) < \tau$ (no pre/dec). **Fix:** one name pair everywhere; never bare τ when both exist.

5.3 E3. Symbol overload on H

H is the published radix index *and* the HBM pool size in $C = \lfloor H/M_{\text{spine}} \rfloor$. **Fix:** radix $\mathcal{H}$ or H_{radix} ; pool capacity $H_{\text{mem}} / M_{\text{pool}}$.

5.4 E4. Lemma naming vs. system name

Lemma uses W_{APP} while the system is PD-Prune/FS. Either rename the lemma quantities to $W_{\text{PD-Prune}}$ or define APP:= the admission rule once.

5.5 E5. Accuracy regressions understated

- Tab. `mt`: Qwen3-8B GSM8K APC 1.000 $\rightarrow$ FS/FS+ 0.938 (one problem lost)—intro claims “matches”.
- Tab. `plus`: APC 0.900 $\rightarrow$ PD-Prune+ 0.863 at $\text{tp}=2$.
- Cross-model FS+ accuracy collapses on some families in commented / appendix tables—if retained, must be discussed as a limitation.

Any-finished 16/16 is not the same as committed-spine winner quality; the paper knows this (Tab. `ablate`) but the abstract/intro sell only the former.

5.6 E6. Prop. 1 is not a proposition as stated

eq:sep-pre–eq:sep-dec are *empirical conditions* on a particular score and residual mix, not theorems. Rename to “Operating-point condition” or “Empirical separator”, and report the score distribution (histogram / quantiles), not four point values.

5.7 E7. Indexing of residuals

Definition 1 (branching step) in Formulation uses $\{\rho_i\}_{i=0}^{k-1}$; Background model text sometimes uses $i = 1..k$. Align.

5.8 E8. DistServe comparison is conceptual, not empirical

The PD figures compare against “stock disaggregation”, but experiments run on vLLM with prefix caching—not DistServe/Splitwise. Either run a real disaggregated baseline or label the comparison as analytical / cost-model only (as Tab. app already is).

6 Experiments that need more detail

Gap	What to add
Score / Loop definition	Spec + ablations: replace Score with random / entropy / reward-model; measure FP/FN of Loop.
Scale	Full GSM8K ($n=1319$) under τ_{pre} gate, not only KV forest; full MATH-500; AIME with confidence intervals.
Real agents	SWE-bench / OpenHands / tool-call traces with natural residual diversity (not planted loops).
True PD stack	Measure X bytes and TTFT on a real disaggregated deployment (or vLLM PD / Mooncake / DistServe-like).
Admission overhead	Microbench Score+hash path latency vs. saved prefill.
Threshold robustness	Sweep $\tau_{\text{pre}}, \tau_{\text{dec}}, t_0, \alpha$ with Pareto (accuracy vs. prefill vs. e2e); not two points.
Baseline fairness	SGLang RadixAttention same hardware; decode pruners with equal any-finished <i>and</i> winner metrics.
Variance	≥ 3 seeds; temperature > 0 for search methods that need it.
Memory vs. latency	Tab. gsm shows FS fan-out <i>slower</i> than APC (140 vs. 87 ms). Main text must own this tradeoff; end-to-end wins need a clear causal attribution (prefill cut vs. D^+ cut vs. stop markers).

7 Notation suggestions

1. Variant table (mandatory).

Name	Alias trunk	Admission
APC / PD-base	no	none
PD-Prune-CoW (FS)	yes	none (all residuals prefilled)
PD-Prune (FS+)	yes	radix prune a_i
PD-Prune +probe	yes	a_i + decode probe (eq. for d_i)

Drop the overloaded name **APP**, or define $\text{APP} \equiv \text{radix prune once}$.

2. **Objects.** Keep trunk x_u , residual ρ_i , spine $x_\star$. Prefer $|\cdot|$ for lengths; avoid mixing L and $|x_u|$ in the same display without $L := |x_u|$.
3. **Actions.** Write the first-match rule as a priority list in text; keep Iverson form in Appendix only.
4. **Work / Transfer.** W_i^{gpu} , X_i^{conn} ; aggregates $W = \sum_i W_i$, $X = \sum_i X_i$.
5. **Scores.** $s_i^{\text{pre}} = \text{Score}_{\text{pre}}(\rho_i)$, $s_i^{\text{dec}} = \text{Score}_{\text{dec}}(y_i)$; never reuse s_i for both.
6. **Pools.** Radix index $\mathcal{H}$; HBM capacity M_{pool} ; slots $C = \lfloor M_{\text{pool}}/M_{\text{spine}} \rfloor$.
7. **Macros.** One macro for the system; one for the “+admission” variant; forbid FS/APP in captions unless defined in that caption.

8 DistServe-style core architecture figure

DistServe’s runtime figure is a **controller + prefill pool + decode pool + KV path**. For PD-Prune we mirror that layout and put the novel gates on the cut:

docs/paper/acl_new/fig/fig_core_arch.tex

Suggested caption:

PD-Prune runtime (DistServe-style). Requests enter a controller that aliases the trunk and runs radix prune before prefill kernels. Only the miss tail of $\mathcal{K}$ crosses the connector. Decode instances run a t_0 probe (and optional ESC/SR/DPTS) and retain the spine after abort.

Use this as the main-paper Figure 1 if `fig_pd` stays as a before/after comparison.

9 Line-level comments (high signal)

- **Abstract:** rewrite entirely; lead with the systems problem in one plain sentence.
- **Intro, “This work”:** too many numbers; move results to Experiments.
- **Prop. host:** good; make it a numbered insight in the intro.
- **Tab. main:** do not win-highlight a single ESC time under a “ESC/SR/DPTS” row—split hosts or report three times without a single green cell.
- **Conclusion:** currently a compressed abstract; add limitations (score dependence; fan-out overhead; venue of applicability).

10 Minimal revision plan (author checklist)

1. Define Score / Loop and release the scorer.
2. Add variant table; purge APP/FS ambiguity.
3. Fix Alg. / H / DraftScore inconsistencies.
4. Scale GSM8K+MATH under the gate; add one real agent workload.
5. Rewrite Abstract + Intro for ACL clarity; cut 30–40% redundancy.
6. Promote `fig_core_arch` to Figure 1; keep `fig_pd` as comparison.
7. Own accuracy drops and APC-faster fan-out latency in the main text.